EE558 – Optical Fiber Comm. Systems
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Homework #2

For the questions below, use the following constants:

Speed of light: c=2.998×10^8 m/s
Electron charge: e=1.602×10^{-19} C
Plank constant: h=6.626×10^{-34} J/Hz

1- A semiconductor laser has length L = 350 μm and index of refraction n = 3.4. At what wavelength near 1.55 μm does this laser satisfy the phase condition for oscillation? What is the spacing, in μm, between these different oscillating modes?

2- The maximum multipath delay in a graded-index (GRIN) fiber is $\frac{\Delta T}{L} = \frac{(n_1 - n_2)^2}{8 \cdot c \cdot n_1}$,

(a) Assume n_1 = 1.5, n_2 = 1.485 and L = 40 km. What is the maximum bit rate that can be transmitted through a multi-mode fiber? (Hint: the bit rate is usually no larger than $\frac{1}{4\Delta T}$)

(b) What if n_1 = 1.49, n_2 = 1.48 and L = 400 km? What is the maximum bit rate?

3- Assume the loss coefficient in an optical fiber is 0.22 dB/km.
(a) Find the optical power of the signal after the transmitter if a signal power of 1 dBm is obtained after transmission distances L = 100 km, 200 km, 300 km, and 400 km?
(b) For the 200 km case, assume the path consists of cascaded fiber cables, each having a length of 20 km, spliced together and each splice has a 0.25 dB loss. Each of the connectors at both ends of the link, which can be used to connect the fiber to the transmitter and receiver, has a loss of 0.5 dB. Find the optical power of the transmitter in order to obtain the same received power as shown in (a).

4- A conventional single-mode-fiber connects Los Angeles to San Francisco. The fiber is 500 km long. An optical source has wavelength λ = 1550 nm and spacing $\Delta\lambda$= 0.1 nm. The dispersion coefficient of the fiber is 16 ps/nm/km.
(a) Find the maximum bit rate based on the parameters given above?
(b) What is the maximum transmission distance (i.e., the same fiber) if we want to transmit the signal at 40 Gb/s and with $\Delta\lambda$= 0.08 nm?

5- If we use a single mode fiber with 17 ps/nm/km dispersion for transmitting two different signals with two different wavelengths – 1550 and 1560 nm respectively –, what is the time delay between those two signals after 100 km?